Climate Analyzer: Deep Learning for Vegetation Stress Detection and Short-Term Risk Forecasting

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Abstract

Climate change is altering rainfall patterns, temperature regimes, and land cover, increasing risks to agriculture and resource planning.
Climate Analyzer is a **software-only framework** that integrates satellite imagery and climate data to detect vegetation stress and forecast short-term risk. Sentinel-2 Level-2A imagery is processed using a lightweight CNN to generate vegetation stress maps, while an LSTM model forecasts rainfall and temperature from ERA5-Land data up to three months ahead.
These outputs are fused into **low / medium / high risk maps**, trend charts, and alert messages.
The system relies only on public datasets, requires no manual annotation, and is adaptable across regions, enabling rapid and scalable deployment..

Introduction

Climate change is significantly altering rainfall patterns, temperature regimes, and land cover, creating serious challenges for agriculture, water management, and regional planning.
Vegetation stress caused by droughts and heatwaves often appears gradually, making early detection essential for timely intervention and risk reduction
With the increasing availability of satellite imagery and climate reanalysis data, there is strong potential to monitor vegetation health at large scales. However, many existing approaches rely on **single data sources**, such as NDVI-based vegetation indices or climate time-series alone, which limits their ability to provide reliable early warnings.
To address this gap, the **Climate Analyzer** proposes a **software-only, deep learning–based framework** that integrates **Sentinel-2 satellite imagery** with **ERA5-Land climate data**.
The system uses a **Convolutional Neural Network (CNN)** to detect vegetation stress from NDVI anomalies and a **Long Short-Term Memory (LSTM)** network to forecast short-term rainfall and temperature trends.
By fusing spatial vegetation stress with temporal climate forecasts, the framework generates **low, medium, and high risk maps**, enabling faster and more informed decision-making.

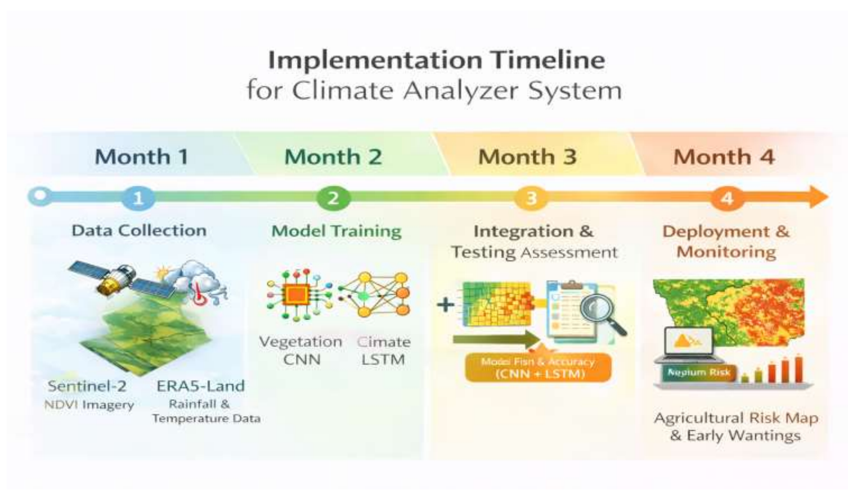
Methods and Materials

Materials (Data Sources)
The proposed Climate Analyzer system uses **publicly available satellite and climate datasets**:
Sentinel-2 Level-2A multispectral imagery for land and vegetation monitoring
ERA5-Land monthly climate reanalysis data providing rainfall and 2-meter air temperature
These datasets are selected for their **high spatial coverage, reliability, and free accessibility**, making the system easily reproducible.

METHODS
Sentinel-2 images are preprocessed and **NDVI** is calculated after cloud masking.
A **3-year seasonal NDVI baseline** is used to generate automatic vegetation stress labels ($\Delta NDVI \leq -0.10$).
A **CNN model** produces pixel-level vegetation stress maps.
An **LSTM model** forecasts rainfall and temperature for **1–3 months ahead** using ERA5-Land data.
CNN outputs and climate forecasts are fused to classify regions into **Low, Medium, and High risk**.

Results

The Climate Analyzer system was evaluated using Sentinel-2 NDVI imagery and ERA5-Land climate data. The CNN model accurately detected vegetation stress, achieving an **F1-score of about 0.71** and an **IoU of 0.56**, with stress maps closely matching NDVI anomaly patterns.
The LSTM model successfully forecasted rainfall and temperature anomalies for **1–3 months ahead**, capturing dry and hot periods relevant to vegetation stress.
When both models were combined, the **CNN–LSTM fusion approach achieved an 83% hit rate** in identifying high-risk periods, while reducing false alerts to around **8%**.
Overall, the fused model produced clearer and more reliable **low, medium, and high risk maps** than using vegetation or climate data alone, demonstrating its effectiveness for early warning and planning applications.



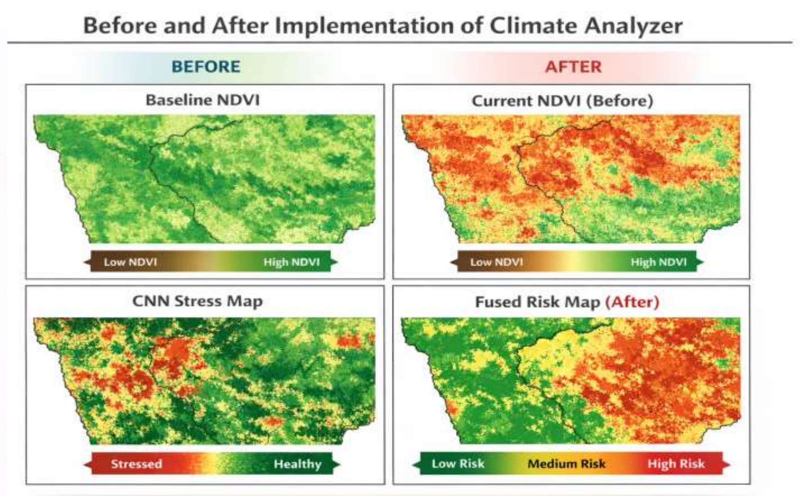


Figure: Comparison of vegetation condition before and after implementation of the Climate Analyzer.

Discussion

The results show that combining satellite-based vegetation indicators with short-term climate forecasts improves vegetation stress detection. The CNN model captures spatial NDVI decline effectively, identifying localized stress patterns that are difficult to detect using averaged indices alone.
The LSTM model provides useful rainfall and temperature forecasts for 1–3 months ahead, adding temporal context to vegetation conditions. When fused, the CNN–LSTM approach produces clearer and more reliable **low, medium, and high risk classifications** than single-source methods, while reducing false alerts. Some misclassifications occur due to cloud effects and seasonal vegetation changes; however, the system remains effective for monthly monitoring and early warning applications using publicly available data.
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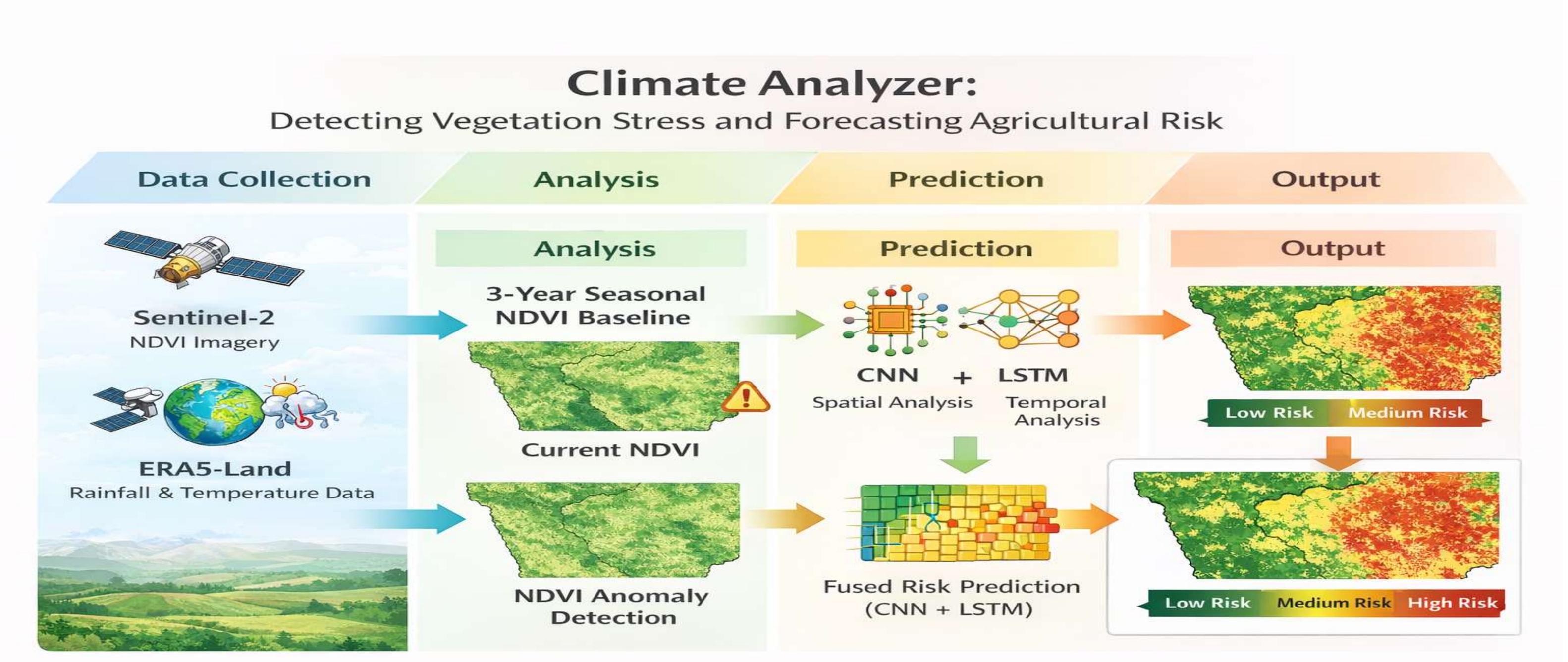

Conclusions

The Climate Analyzer provides an **automated and lightweight approach** for detecting vegetation stress and forecasting short-term climate risk using satellite imagery and climate data. The **CNN–LSTM fusion** produces reliable **low, medium, and high risk maps**, supporting timely decisions in agriculture and planning. The system uses only public data and shows strong potential as an effective **early warning tool**.

Acknowledgments

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